ME 264 Mechanisms Synthesis & Analysis I

Units 2 Graphical Dimensional Synthesis of Linkages

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Linkage Transformation

- Formalized Synthesis techniques are useful for designing linkages of particular *DOF.*
- If the Synthesis techniques rules are relaxed, the arbitrary constraint, which restrict joints to only revolute joints, may be utilized to transform basic linkages to a wider variety of mechanisms with even greater usefulness.
- There are several transformation techniques or rules that we can apply to planar kinematic chains.

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Transformation Techniques/Rules

1. Revolute joints in any loop can be replaced by prismatic joints with no change in *DOF* of the mechanism, provided that at least two revolute joints remain in the loop.

2. Any full joint can be replaced by a half joint, but this will increase the *DOF* by one.

3. Removal of a link will reduce the *DOF* by one.

3

Transformation Techniques/Rules

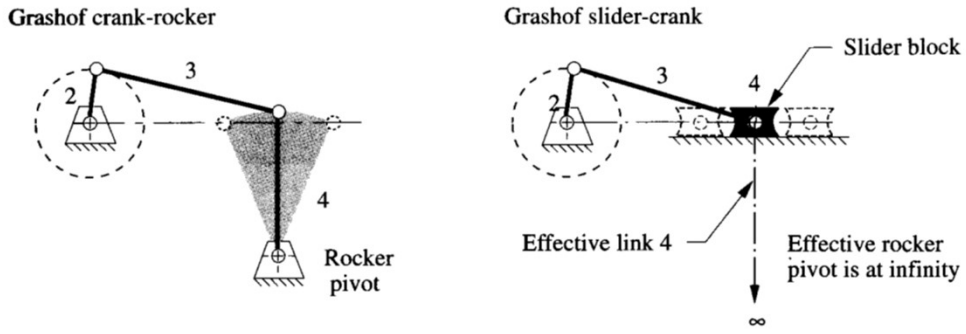
4. Combination of rules 2 and 3 will keep the original *DOF* unchanged.

5. Any ternary (3-node) or higher-order link can be partially "shrunk" to a lower-order link by coalescing nodes.
   - This will create a multiple joint but will not change the *DOF* of the mechanism.
   - Complete shrinkage of a higher-order link is equivalent to its removal. A multiple joint will be created, and the *DOF* will be reduced.

4

Transformation Techniques/Rules

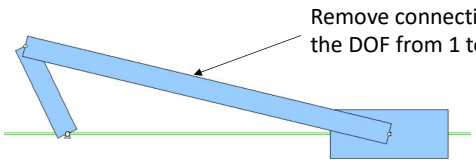
Application of Rule 1 to transfer a four-bar rocker into a four-bar crank-slider mechanism



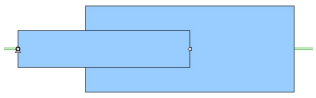
(a) Transforming a fourbar crank-rocker to a slider-crank

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Transformation Techniques/Rules

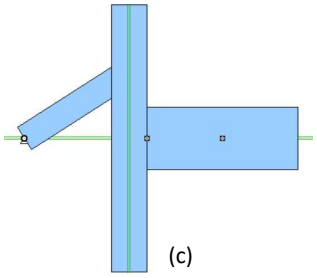


(a)



(b)
Resulting linkage (b) has DOF =0

- Replace J1 joint between crank and slider with J_2.
- This will increase DOF from 0 to 1 (Rule 2)
- Combination of rules 2 and 3 transform figure (a) to figure (c) with same DOF 1 and number of links reduce to 3.



(c)
Resulting linkage (c) has DOF =1

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SYNTHESIS

QUALITATIVE SYNTHESIS

- It is *the creation of potential solutions in the absence of a well-defined algorithm which configures or predicts the solution.*
- Since most real design problems will have many more unknown variables than you will have equations to describe the system's behavior, you cannot simply solve the equations to get a solution.
- Nevertheless one must work in this fuzzy context to create a potential solution and to also judge its quality**.**

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TYPE SYNTHESIS

- It refers to *the definition of the proper type of mechanism "best" suited to the problem. This* is a form of qualitative synthesis.
- It is perhaps the most difficult task for the student as it requires some experience and knowledge of the various types of mechanisms which exist and may be feasible from a performance and manufacturing standpoint.

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QUANTITATIVE SNTHESIS,OR ANALYTICALSYNTHESIS

- It refers to the generation of one or more solutions of a particular type which are suitable to the problem, and more importantly, one for which there is a synthesis algorithm defined.
- This type of synthesis can be quantified with some set of equations, which will give a numerical answer. Whether that answer is a good or suitable one is still a matter for the judgment of the designer and requires analysis and iteration to optimize the design.
- Except for very simple cases, a Computer Aided Engineering [also called Computer Aided Design (CAD)] tool is needed to do quantitative synthesis.

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DIMENSIONALSYNTHESIS

- DIMENSIONALSYNTHESIS of a linkage *is the determination of the proportions (lengths) of the links necessary to accomplish the desired motions.*
- It is a form of quantitative synthesis if an algorithm is defined for the particular problem, but can also be a form of qualitative synthesis if there are more variables than equations. The latter situation is more common for linkages. (Dimensional synthesis of cams is quantitative.)
- Dimensional synthesis assumes that, through *type synthesis,* one has already determined that a linkage (or a cam) is the most appropriate solution to the problem. This course focuses on dimensional synthesis and will go into type synthesis and quantitative synthesis.

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Types of Dimensional Synthesis

1. **Function Generation**
2. **Path Generation**
3. **Motion Generation**

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FUNCTION GENERATION

- It is defined as *the correlation of an input motion with an output motion in a mechanism.*
- A function generator is conceptually a "black box" which delivers some predictable output in response to a known input.
- Historically, before the advent of electronic computers, mechanical function generators found wide application. They are, in fact, mechanical analog computers.
- The development of inexpensive digital electronic microcomputers for control systems coupled with the availability of compact servo and stepper motors has reduced the demand for these mechanical function generator linkage devices.
- Many such applications can now be served more economically and efficiently with electromechanical devices. Moreover, the computer-controlled electromechanical function generator is programmable, allowing rapid modification of the function generated as demands change. For this reason, linkage function generators will not be emphasized, except cams which are beyond the scope of ME 264.

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PATH GENERATION

- It is defined as the control of a point in the plane such that it follows some prescribed path.
- In path generation, no attempt is made in path generation to control the orientation of the link which contains the point of interest.
- However, it is common for the timing of the arrival of the point at particular locations along the path to be defined. This is path generation with prescribed timing.

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MOTION GENERATION

- It is defined as the control of a line in the plane such that it assumes some prescribed set of sequential positions
- Unlike path generation, here orientation of the link containing the line is important. This is a more general problem than path generation, and in fact, path generation is a subset of motion generation.

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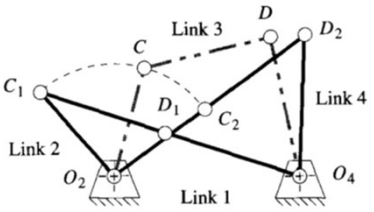
LIMITING CONDITIONS

- The manual, graphical, dimensional synthesis techniques are reasonably rapid means to obtain a trial solution to a motion control problem.
- Once a potential solution is found, it must be evaluated for its quality. There are many criteria which may be applied.
- One need to check that the linkage can in fact reach all of the specified design positions without encountering a limit or **TOGGLE** position, also called a stationary configuration.

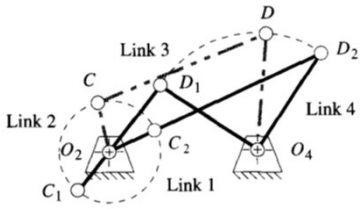
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Toggle Positions

- The toggle positions are determined by the colinearity of two of the moving links.
- A fourbar double- or triple-rocker mechanism will have at least two of these toggle positions in which the linkage assumes a triangular configuration.
- When in a triangular (toggle) position, it will not allow further input motion in one direction from one of its rocker links (either of link 2 from position C_1D_1 or link 4 from position C_2D_2.
- A Grashof fourbar crank-rocker linkage will also assume two toggle positions, when the shortest link (crank 0_2C) is colinear with the coupler CD (link 3), either extended colinear ($0_2C_2D_2$) or overlapping colinear ($0_2C_1D_1$).
- These toggle positions in Grashof four crank-rocker linakgae also define the limits of motion of the driven rocker (link 4), at which its angular velocity will go through zero.



Toggle Positions of fourbar non-Grashof

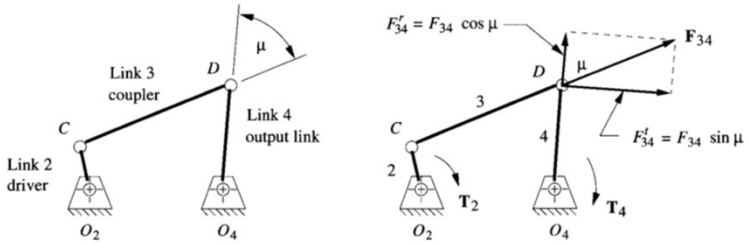


Toggle Positions of fourbar Grashof Crank-Rocker

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TRANSMISSION ANGLE

- It is useful test that can be very quickly applied to a linkage design to judge its quality is the measurement of its transmission angle. This can be done analytically, graphically on the drawing board, or with the cardboard model for a rough approximation.
- The **transmission angle** is defined as the angle between the output link and the coupler. It is usually taken as the absolute value of the acute angle of the pair of angles at the intersection of the two links and varies continuously from some minimum to some maximum value as the linkage goes through its range of motion. It is a measure of the quality of force and velocity transmission at the joint.



(a) Linkage transmission angle μ (b) Static forces at a linkage joint

GRASHOF CONDITION FOR ROTATABILITY

The **Grashof** condition is a very simple relationship which predicts the *rotation behavior* or **rotatability** of a fourbar linkage's inversions based only on the link lengths. **Grashof** condition for **rotatability** is stated as

If ***S+L ≤ Q+P,***

Then the linkage is **Grashof** and at least one link will be capable of making a full revolution with respect to the ground plane.

Where S = *length of shortest link, L = length of longest link, P* = *length of one remaining link,* Q = *length of other remaining link.*

A fourbar linkage that meets the **Grashof condition** is called Grashof linkge or simply **Grashof.**

DIMENSIONAL SYNTHESIS

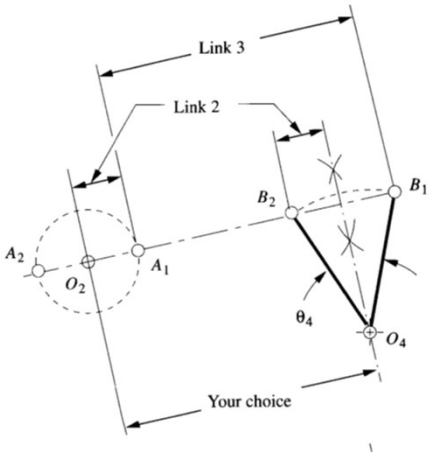
- **Two-Position Synthesis**
- **Three-Position Synthesis**

Two-Position Synthesis

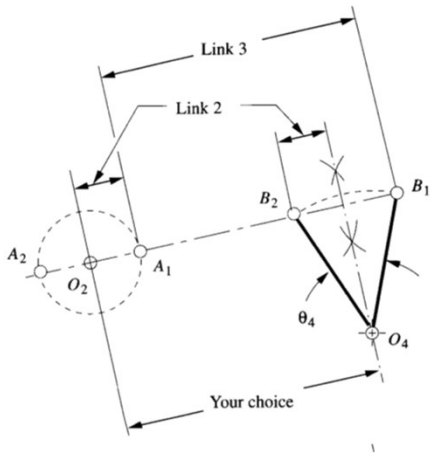
- *Problem 1:* Design a fourbar Grashof crank-rocker to give 45° of rocker rotation with equal time forward and back, from a constant speed motor input.
- *Two Positions with angular Displacement (Function Generation)*

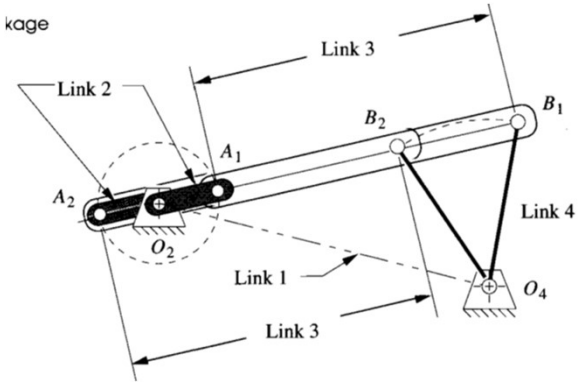
Two Positions with angular Displacement (Function Generation): Solution Steps

1. Draw the output link *O4B* in both extreme positions, *B1* and *B2* in any convenient location, such that the desired angle of motion θ is subtended.
2. Draw the chord *B1B2* and extend it in any convenient direction.
3. Select a convenient point O2 on line *B1B2* extended.
4. Bisect line segment *B1B2 ,* and draw a circle of that radius about *02.*
5. Label the two intersections of the circle and *B1B2* extended, *A1* and *A2.*
6. Measure the length of the coupler as *A 1* to *B1* or *A2* to *B2.*
7. Measure ground length I, crank length 2, and rocker length 4.
8. Check the Grashof condition. If non-Grashof, redo steps 3 to 8 with O2 further from *04.*

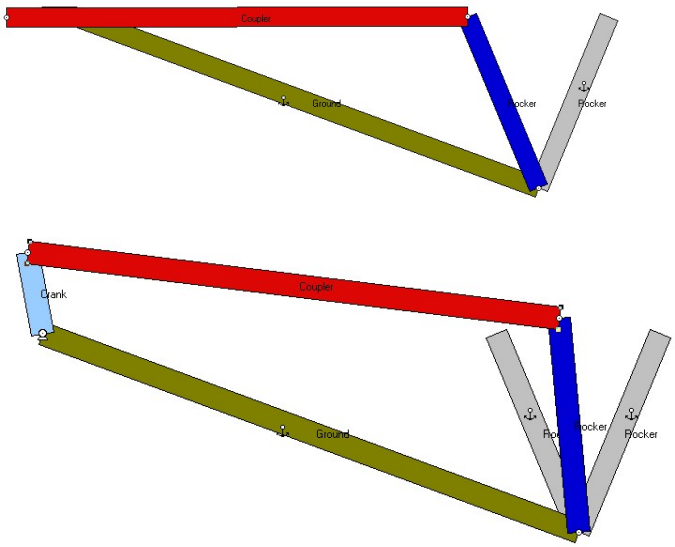


Output





Working Model Simulation



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Assignment 1

1. Design a four-bar Grashof crank-rocker to give 60° of rocker rotation with equal time forward and back, from a constant speed motor input. Use rocker of length 40 cm.

   Outputs:
   1. Hand construction of linkage
   2. Working Model 2D simulation of linkage
   3. Cardboard Model of linkage

Assignment 2

1. Design a four-bar Grashof crank-rocker to give 90° of rocker rotation with equal time forward and back, from a constant speed motor input. Use rocker of length 35 cm.

   Outputs:
   1. Hand construction of linkage
   2. Cardboard Model of linkage

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Solution Steps: Problem 2

Rocker Output - Two Positions with Complex Displacement. (Motion Generation)

Problem: Design a fourbar linkage to move link CD from position C_1D_1 to C_2D_2.

Solution: (see Figure 3-5)

1 Draw the link CD in its two desired positions, C_1D_1 and C_2D_2, in the plane as shown.

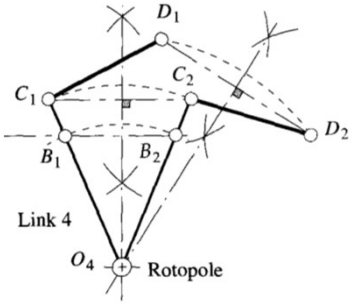
2 Draw construction lines from point C_1 to C_2 and from point D_1 to D_2.

3 Bisect line C_1C_2 and line D_1D_2 and extend their perpendicular bisectors to intersect at O_4. Their intersection is the **rotopole**.

4 Select a convenient radius and draw an arc about the rotopole to intersect both lines O_4C_1 and O_4C_2. Label the intersections B_1 and B_2.

5 Do steps 2 to 8 of Example 3-1 above to complete the linkage.

6 Make a cardboard model of the linkage and articulate it to check its function and its transmission angles.



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1 Draw the link CD in its two desired positions, C_1D_1 and C_2D_2, in the plane as shown.

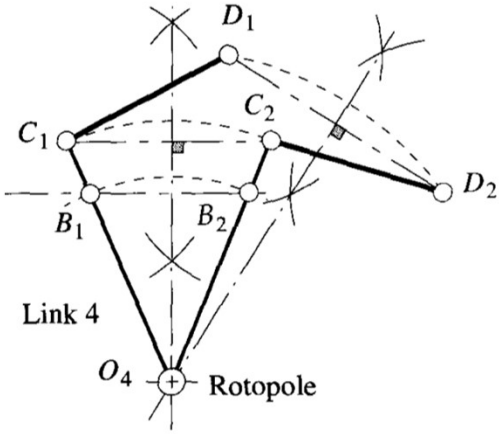
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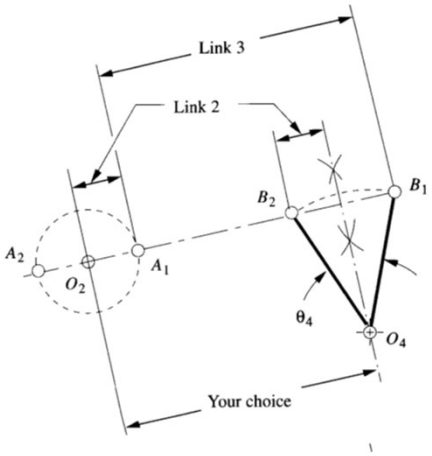
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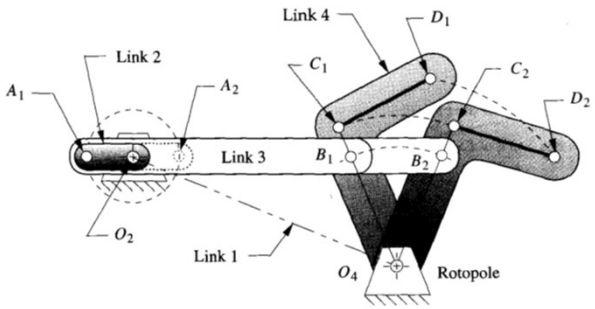


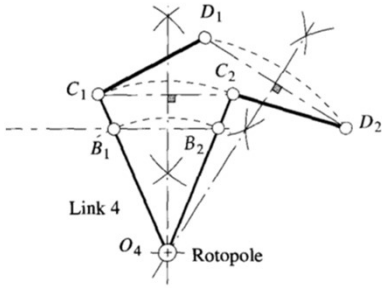
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How to Add Driver Dyad

1. Draw the output link *O4* in both extreme positions, *B1* and *B2* in any convenient location, such that the desired angle of motion 45 is subtended.
2. Draw the chord *B1B2* and extend it in any convenient direction.
3. Select a convenient point O2 on line *B1B2* extended.
4. Bisect line segment *B1B2 ,* and draw a circle of that radius about *02.*
5. Label the two intersections of the circle and *B1B2* extended, *A1* and *A2.*
6. Measure the length of the coupler as *A 1* to *B1* or *A2* to *B2.*
7. Measure ground length I, crank length 2, and rocker length 4.
8. Check the Grashof condition. If non-Grashof, redo steps 3 to 8 with O2 further from *04.*



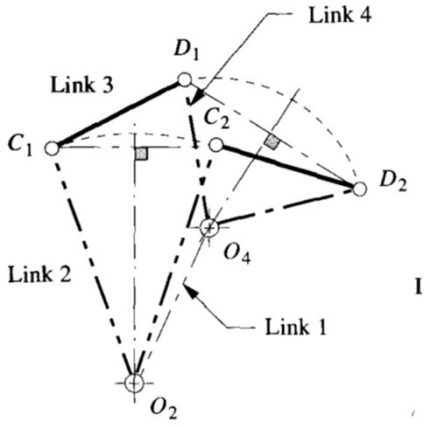




- *Problem 3:* Design a fourbar to move link CD from position C_1D_1 to C_2D_2.
- *Two Positions with Complex Displacement (Motion Generation + Coupler output)*

Solution Steps: Problem 3

1. Draw the link CD in its two desired positions, C1D1 and *C2D2,* in the plane as shown.
2. Draw construction lines from point C1 to C2 and from point D1 to *D2.*
3. Bisect line C1C2 and line *D)D2* and extend the perpendicular bisectors in convenient directions. The rotopole will not be used in this solution.



Solution Steps: Problem 3

4 Select any convenient point on each bisector as the fixed pivots O_2 and O_4, respectively.

5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.

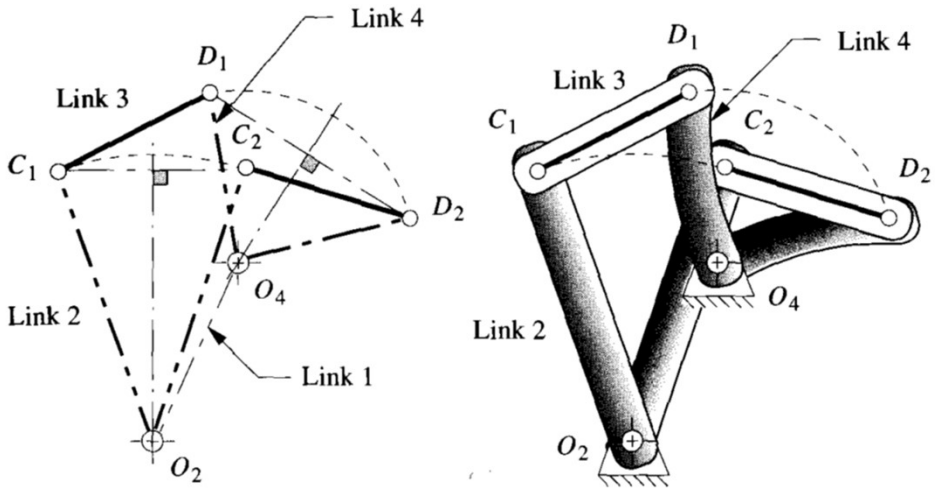
6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.

7 Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.

8 Construct a cardboard model and check its function to be sure it can get from the initial to final position without encountering any limit (toggle) positions.

9 Check transmission angles.

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Adding a Dyad (Twobar Chain) to Control Motion in Example 3-3.

Problem: Design a **dyad** to control and limit the extremes of motion of the linkage in Example 3-3 to its two design positions.

Solution:

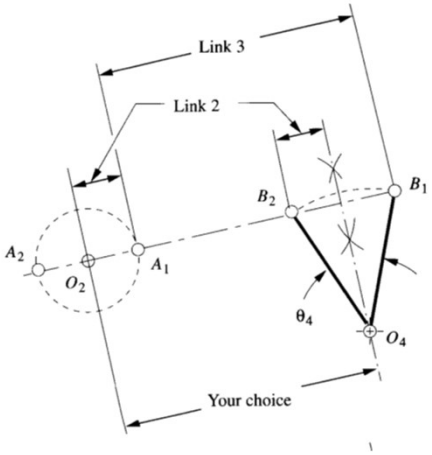
1 Select a convenient point on link 2 of the linkage designed in Example 3-3. Note that it need not be on the line O_2C_1. Label this point B_1.

2 Draw an arc about center O_2 through B_1 to intersect the corresponding line O_2B_2 in the second position of link 2. Label this point B_2. The chord B_1B_2 provides us with the same problem as in Example 3-1.

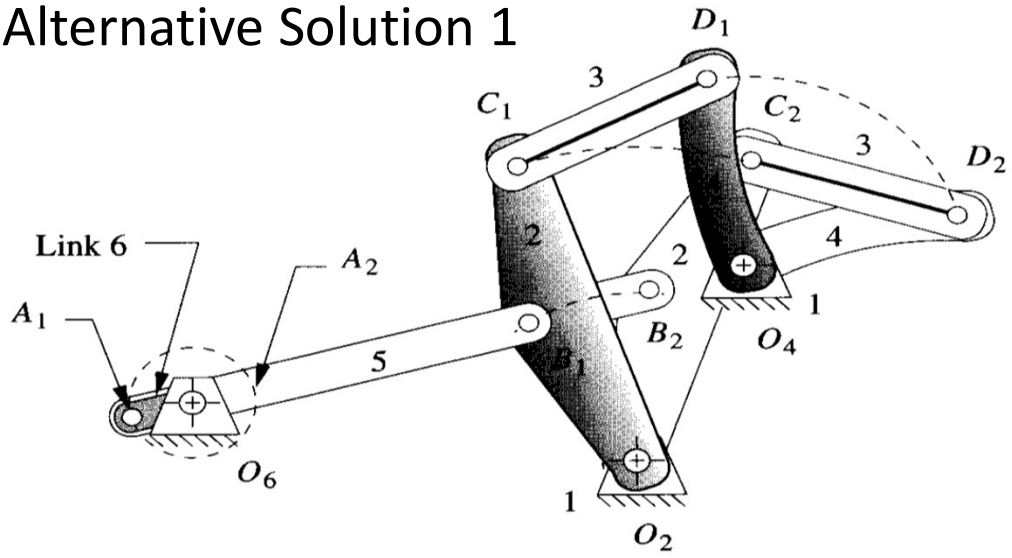
3 Do steps 2 to 9 of Example 3-1 to complete the linkage, except add links 5 and 6 and center O_6 rather than links 2 and 3 and center O_2. Link 6 will be the driver crank. The fourbar subchain of links O_6, A_1, B_1, O_2 must be a Grashof crank-rocker.

How to Add Driver Dyad

1. Draw the output link *O4* in both extreme positions, *B1* and *B2* in any convenient location, such that the desired angle of motion 45 is subtended.
2. Draw the chord *B1B2* and extend it in any convenient direction.
3. Select a convenient point O2 on line *B1B2* extended.
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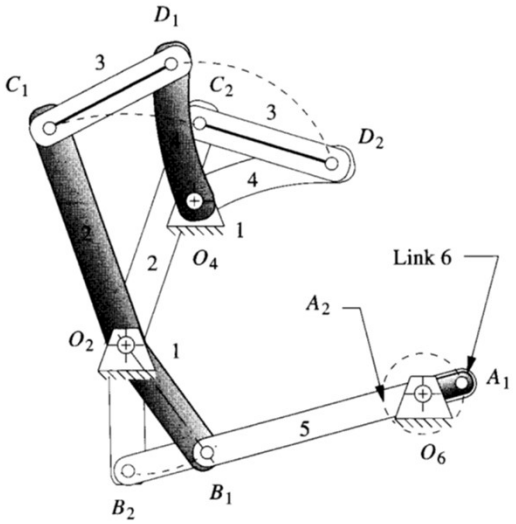
Alternative Solution 1



(*b*) The completed Watt's sixbar linkage with motor at O_6

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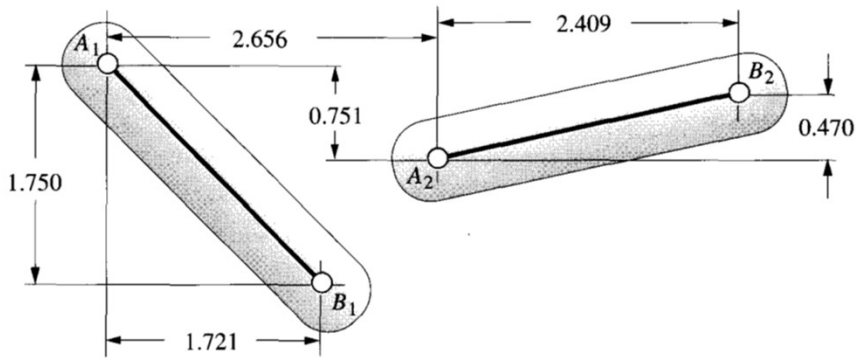
Alternative Solution 2



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Assignment 3
Design a fourbar mechanism to give the two positions shown to give rocker motion with no quick-return. Build a cardboard model and determine the minimum transmission angle.

Assignment 4
Design a fourbar mechanism to give the two positions shown to give coupler motion with no quick-return. Build a cardboard model and determine the minimum transmission angle.

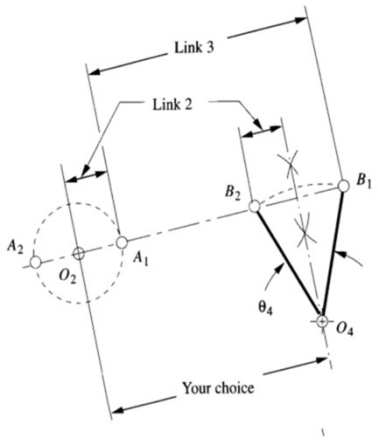


QUICK-RETURN MECHANISMS

- Many machine design applications have a need for a difference in average velocity between their "forward" and "return" strokes.
- Typically some external work is being done by the linkage on the forward stroke, and the return stroke needs to be accomplished as rapidly as possible so that a maximum of time will be available for the working stroke.

Fourbar Quick-Return

- It is a crank-rocker which gives two positions of the rocker with unequal equal time for the forward stroke and the return stroke.
- Equal time for the forward stroke and the return stroke crank-rocker linkage is called a *non-quick-return* linkage.
- The reason for its non quick-return state is the positioning of the crank center 0_2 on the chord B_1B_2 extended. This results in equal angles of 180 degrees being swept out by the crank as it drives the rocker from one extreme (toggle position) to the other.
- If the crank is rotating at constant angular velocity, as it will tend to when motor driven, then each 180 degree sweep, forward and back, will take the same time interval.



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Fourbar Quick-Return

$$TR = \frac{\alpha}{\beta} \qquad (1)$$

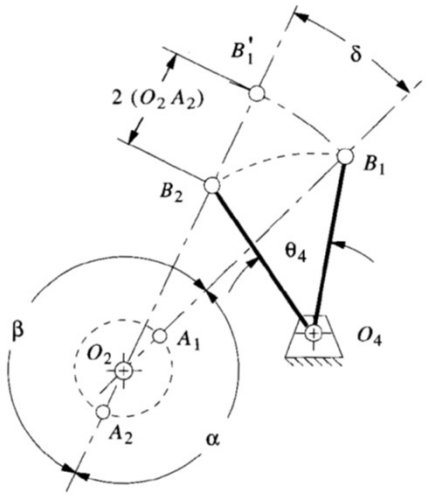
$$\alpha + \beta = 360 \qquad (2)$$

Where TR= timing ratio, α *and* β are timing angles. Solving (1) and (2), we have

$$\beta = \frac{360}{1 + TR} \qquad (3)$$

Calculate α = 360- β and δ

$$\delta = |180 - \beta| = |180 - \alpha| \qquad (4)$$



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Timing Ratio Example

TR = 1.25 $\qquad \alpha + \beta = 360$

TR = 1.25 $= \frac{\alpha}{\beta} \qquad \alpha = 1.25\beta$

$$1.25\beta + \beta = 2.25\beta = 360$$

$$\therefore \beta = \frac{360}{2.25} \quad \beta = 160$$

$$\alpha = 360 - 160 \quad \alpha = 200$$

$$\delta = |180 - \alpha| = |180 - 160|$$

$$\delta = 20$$

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Steps for construction

1 Draw the output link O_4B in both extreme positions, in any convenient location, such that the desired angle of motion, θ_4, is subtended.

2 Calculate α, β, and δ using equations 3.1 and 3.2. In this example, $\alpha = 160°$, $\beta = 200°$, $\delta = 20°$

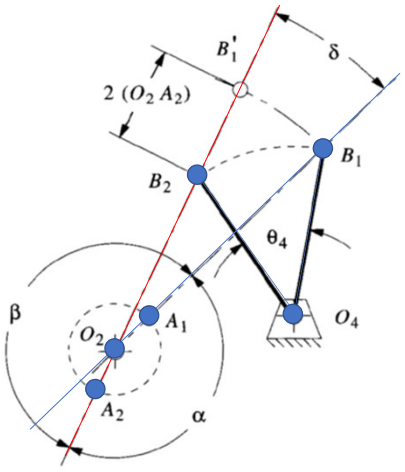
3 Draw a construction line through point B_1 at any convenient angle.

4 Draw a construction line through point B_2 at angle δ from the first line.

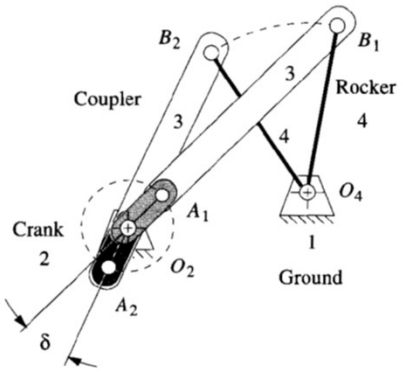
5 Label the intersection of the two construction lines O_2.

6 The line O_2O_4 now defines the ground link.

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(a) Construction of a quick-return Grashof crank-rocker



(b) The finished linkage in its two toggle positions

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Steps for construction

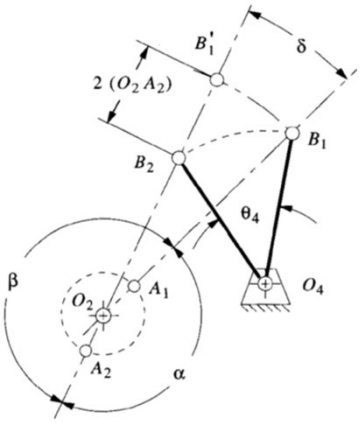
7 Calculate the lengths of crank and coupler by measuring O_2B_1 and O_2B_2 and solve simultaneously:

$$\text{Coupler} + \text{crank} = O_2B_1$$

$$\text{Coupler} - \text{crank} = O_2B_2$$

or you can construct the crank length by swinging an arc centered at O_2 from B_1 to cut line O_2B_2 extended. Label that intersection B_1'. The line B_2B_1' is twice the crank length. Bisect this line segment to measure crank length O_2A_1.

8 Calculate the Grashof condition. If non-Grashof, repeat steps 3 to 8 with O_2 further from O_4.



$$Coupler = \frac{O_2B_1 + O_2B_2}{2} \qquad (5)$$

$$Crank = \frac{O_2B_1 - O_2B_2}{2} \qquad (6)$$

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CRANK-SLIDER QUICK-RETURN A commonly used mechanism, capable of large time ratios is shown in Figure 3-14. It is often used in metal shaper machines to provide

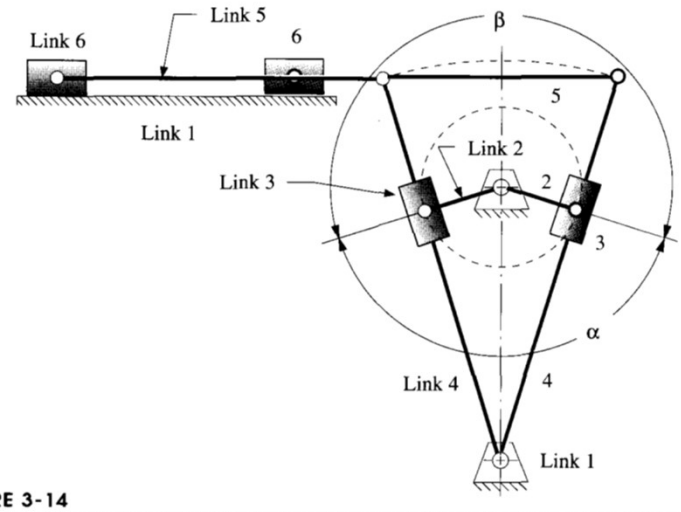


FIGURE 3-14
Crank-shaper quick-return mechanism

Assignment 5

1. Design a four-bar Grashof crank-rocker to give 60° of rocker rotation with quick-return of timing ratio 1:1.25 forward and back, from a constant speed motor input. Use rocker of length 40 cm.

   Outputs:

   1. Hand construction of linkage
   2. Working Model 2D simulation of linkage
   3. Cardboard Model of linkage

Three-Position Synthesis with Specified Moving Pivots

Problem

Design a fourbar linkage to move the link CD shown from position C1D1 to C2D2 and then to C3D3 using the moving pivots at C and D. Find the fixed pivot locations

Output

Motion Generation+ Coupler Output with complex displacement

Steps for construction

1 Draw link CD in its three design positions C_1D_1, C_2D_2, C_3D_3 in the plane as shown.

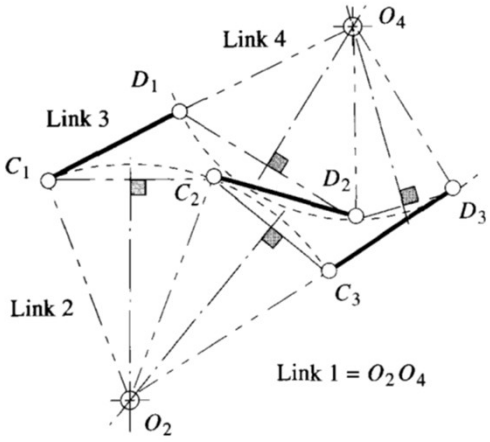
2 Draw construction lines from point C_1 to C_2 and from point C_2 to C_3.

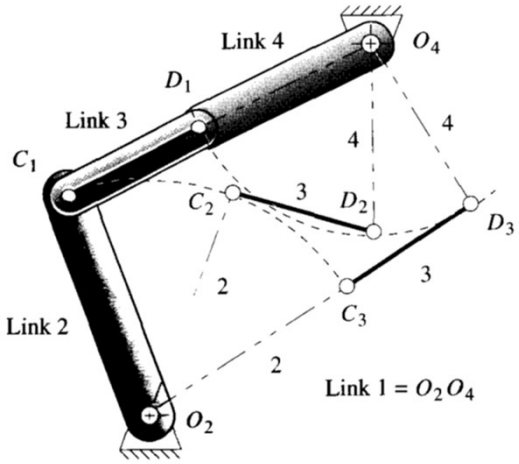
3 Bisect line C_1C_2 and line C_2C_3 and extend their perpendicular bisectors until they intersect. Label their intersection O_2.

4 Repeat steps 2 and 3 for lines D_1D_2 and D_2D_3. Label the intersection O_4.

5 Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.

Steps for construction





Steps for construction

6 Line C_1D_1 is link 3. Line O_2O_4 is link 1.

7 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.

8 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.

9 Construct a driver dyad according to the method in Example 3-4 using an extension of link 3 to attach the dyad.

Three-Position Synthesis with Alternative Moving Pivots

Problem

Design a fourbar linkage to move the link CD shown from position C1D1 to C2D2 and then to C3D3 using different moving pivots than C and D. Find the fixed pivot locations

Output

Motion Generation+ Coupler Output with complex displacement

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Steps for construction

1 Draw the link CD in its three desired positions C_1D_1, C_2D_2, C_3D_3, in the plane as done in Example 3-5.

2 Define new attachment points E_1 and F_1 which have a fixed relationship between C_1D_1 and E_1F_1 within the link. Now use E_1F_1 to define the three positions of the link.

3 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3.

4 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection O_2.

5 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3. Label the intersection O_4 .

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Steps for construction

6 Connect O_2 with E_1 and call it link 2. Connect O_4 with F_1 and call it link 4.

7 Line E_1F_1 is link 3. Line O_2O_4 is link 1.

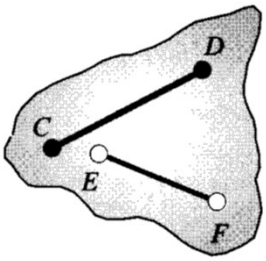
8 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.

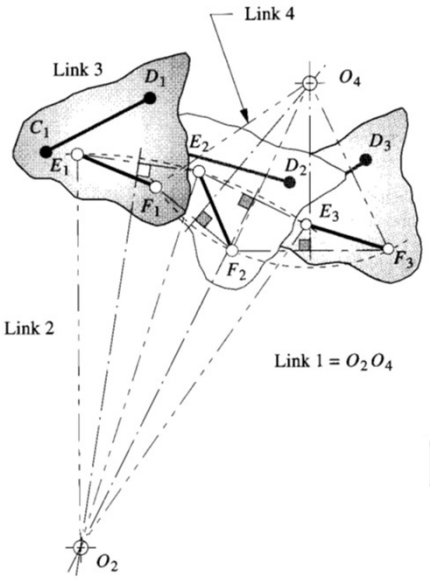
9 Construct a cardboard model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions. If not, change locations of points E and F and repeat steps 3 to 9.

10 Construct a driver dyad acting on link 2 according to the method in Example 3-4.

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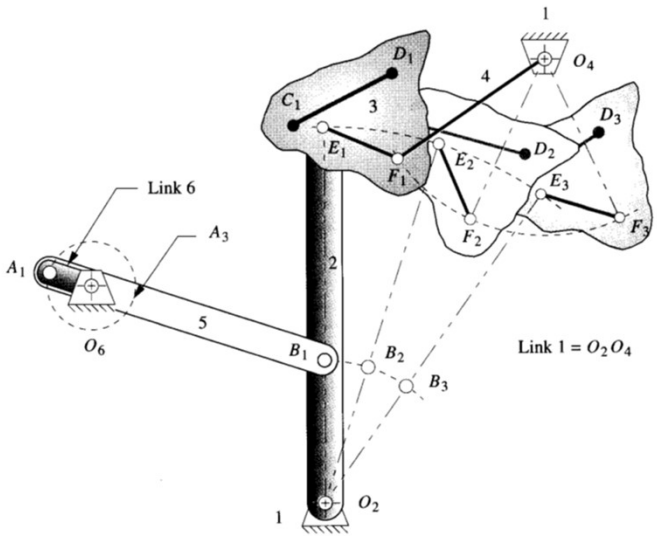
Alternate attachment points





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Complete Linkage with a dyad



Three-Position Synthesis with Specified Moving Fixed Pivots-Inverting the three-position Motion synthesis problem

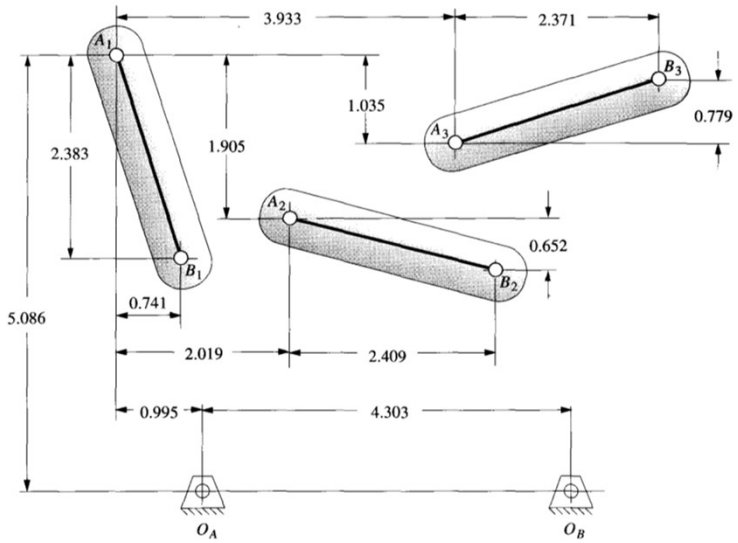
Problem

Design a fourbar linkage to move the link CD shown from position C1D1 to C2D2 and then to C3D3 using the specified fixed pivots O2 and O4. Find the fixed pivot locations

Output

Motion Generation+ Coupler Output with complex displacement

Example



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Phase 1
Steps for construction

1 Draw the link CD in its three desired positions C_1D_1, C_2D_2, C_3D_3, in the plane, as was done in Example 3-5 and as shown in Figure 3-10a.

2 Draw the ground link O_2O_4 in its desired position in the plane with respect to the first coupler position C_1D_1 as shown in Figure 3-10a.

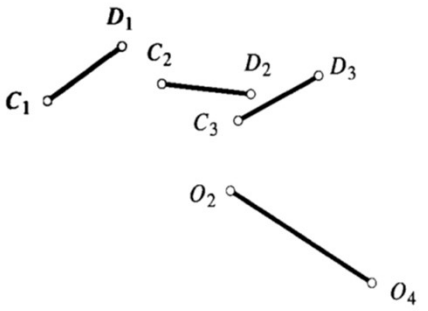
3 Draw construction arcs from point C_2 to O_2 and from point D_2 to O_2 whose radii define the sides of triangle $C_2O_2D_2$. This defines the relationship of the fixed pivot O_2 to the coupler line CD in the second coupler position as shown in Figure 3-10b.

4 Draw construction arcs from point C_2 to O_4 and from point D_2 to O_4 to define the triangle $C_2O_4D_2$. This defines the relationship of the fixed pivot O_4 to the coupler line CD in the second coupler position as shown in Figure 3-10b.

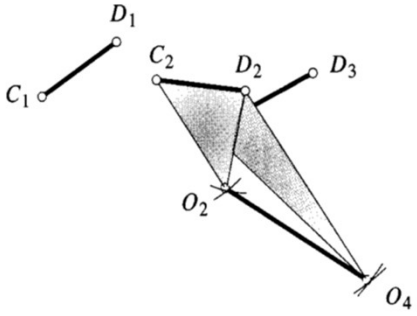
5 Now transfer this relationship back to the first coupler position C_1D_1 so that the ground plane position $O_2'O_4'$ bears the same relationship to C_1D_1 as O_2O_4 bore to the second coupler po-

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Inverting the three-position Transferring of 2nd Position



(a) Original coupler three-position problem with specified pivots



(b) Position of the ground plane relative to the second coupler position

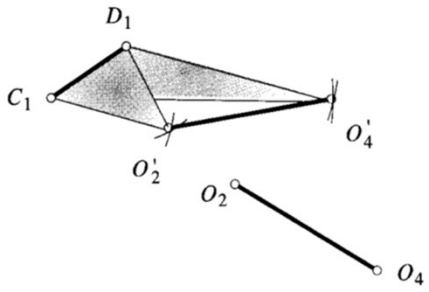
Steps for construction

sition C_2D_2. By doing this, we have pretended that the ground plane moved from O_2O_4 to $O_2'O_4'$ instead of the coupler moving from C_1D_1 to C_2D_2. That is, we have *inverted* the problem.

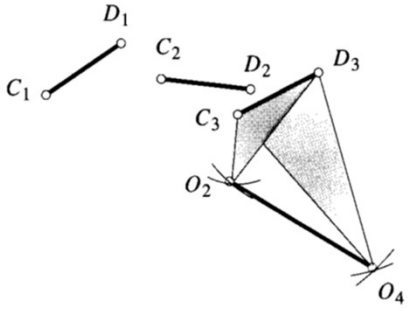
6 Repeat the process for the third coupler position as shown in Figure 3-10d and transfer the third relative ground link position to the first, or reference, position as shown in Figure 3-10e.

7 The three inverted positions of the ground plane which correspond to the three desired coupler positions are labeled O_2O_4, $O_2'O_4'$, and $O_2''O_4''$ and have also been renamed E_1F_1, E_2F_2, and E_3F_3 as shown in Figure 3-10f. These correspond to the three coupler positions shown in Figure 3-10a. Note that the original three lines C_1D_1, C_2D_2, and C_3D_3 are not now needed for the linkage synthesis.

Inverting the three-position Transferring of 3rd Position



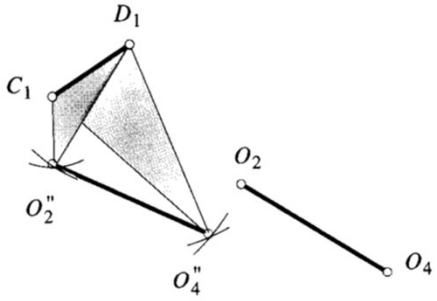
(c) Transferring second ground plane position to reference location at first position



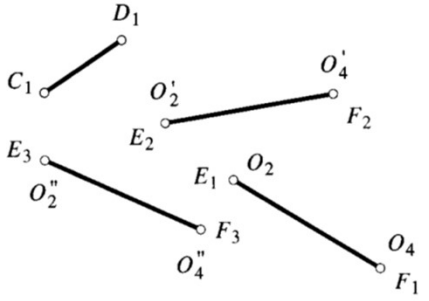
(d) Position of the ground plane relative to the third coupler position

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Three Inverted position after transferring of 2nd and 3rd Positions



(e) Transferring third ground plane position to reference location at first position



(f) The three inverted positions of the ground plane corresponding to the original coupler position

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Using the inverted ground link positions E_1F_1, E_2F_2, and E_3F_3 found in Example 3-7, find the fixed pivots for that inverted motion, then reinvert the resulting linkage to create the moving pivots for the three positions of coupler CD which use the selected fixed pivots O_2 and O_4 as shown in Figure 3-10a (see also Figure 3-11).

1 Start with the inverted three positions in the plane as shown in Figures 3-10f and 3-11a. Lines E_1F_1, E_2F_2, and E_3F_3 define the three positions of the inverted link to be moved.

2 Draw construction lines from point E_1 to E_2 and from point E_2 to E_3.

3 Bisect line E_1E_2 and line E_2E_3 and extend the perpendicular bisectors until they intersect. Label the intersection G.

4 Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3. Label the intersection H.

5 Connect G with E_1 and call it link 2. Connect H with F_1 and call it link 4. See Figure 3-11b.

6 In this inverted linkage, line E_1F_1 is the coupler, link 3. Line GH is the "ground" link 1.

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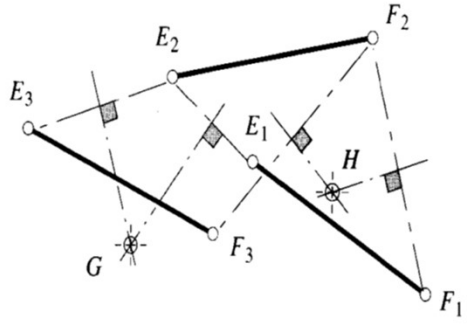
7 We must now reinvert the linkage to return to the original arrangement. Line E_1F_1 is really the ground link O_2O_4, and GH is really the coupler. Figure 3-11c shows the reinversion of the linkage in which points G and H are now the moving pivots on the coupler and E_1F_1 has resumed its real identity as ground link O_2O_4. (See Figure 3-10e.)

8 Figure 3-11d reintroduces the original line C_1D_1 in its correct relationship to line O_2O_4 at the initial position as shown in the original problem statement in Figure 3-10a. This forms the required coupler plane and defines a minimal shape of link 3.

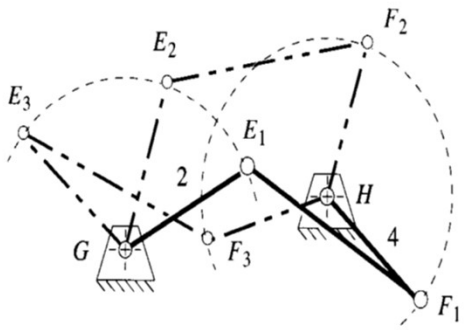
9 The angular motions required to reach the second and third positions of line CD shown in Figure 3-11e are the same as those defined in Figure 3-11b for the linkage inversion. The angle F_1HF_2 in Figure 3-11b is the same as angle $H_1O_4H_2$ in Figure 3-11e and F_2HF_3 is the same as angle $H_2O_4H_3$. The angular excursions of link 2 retain the same relationship between Figure 3-11b and e as well. The angular motions of links 2 and 4 are the same for both inversions as the link excursions are relative to one another.

10 Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case provided that the linkage has mobility among all three positions. This solution is a non-Grashof linkage.

64

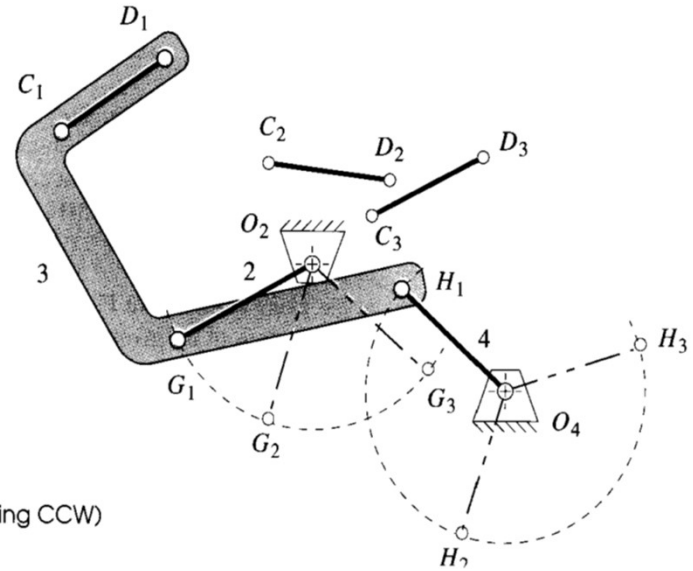


(a) Construction to find rotopoles G and H



(b) The correct inversion of desired linkage

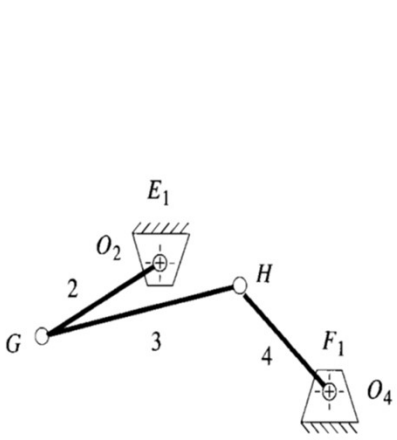
65



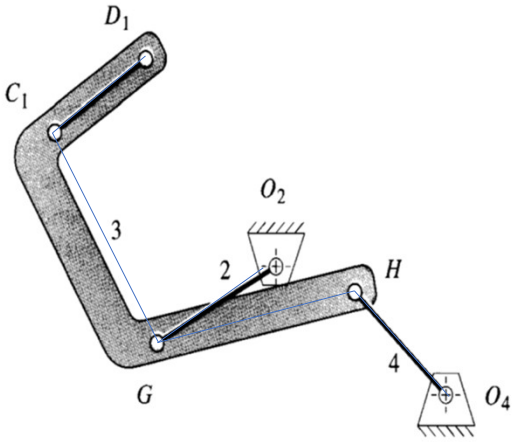
(e) The three positions (link 4 driving CCW)

66

Add driver dyad to limit motion



(c) Reinvert to obtain the result

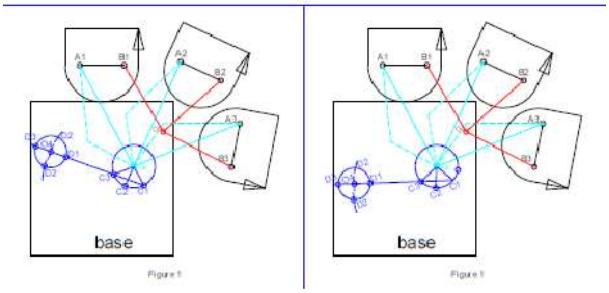


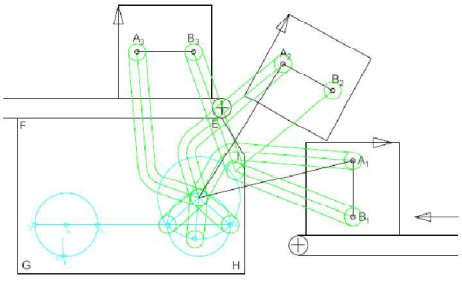
(d) Re-place line *CD* on link 3

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Realigning Dyad due to space constrain

- At the toggles positions, check if any output link pass over a fixed pivot.
- At the toggle position where it may pass over/hit a fixed pivot, bend the links around the ground pivots.
- Draw the bent links at all positions

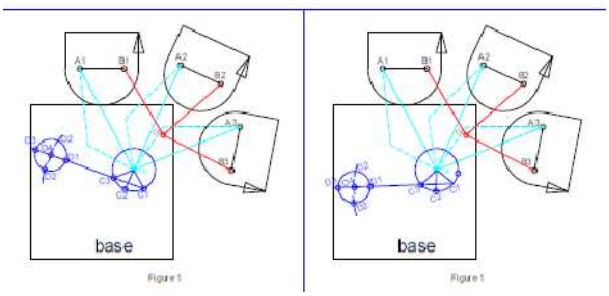


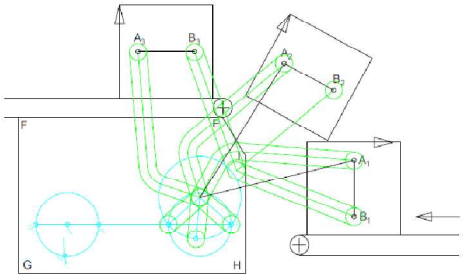


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Adjusting Linkage to Avoid Obstacles

1. Check that links connected to coupler link can move smoothly from one to toggle position to another with hitting a fixed pivot.
2. Draw a line from the fixed pivot of the rocker output of the dyad and perpendicular to the dyad at the two toggle positions.
3. Measure the angle of displacement from one toggle position to the other, and draw two lines, each at ½ of the measured angle on each side of the perpendicular line. Extend the lines to intersect the circle of the dyad at C1 and C2 (or C3 for 3 position synthesis)

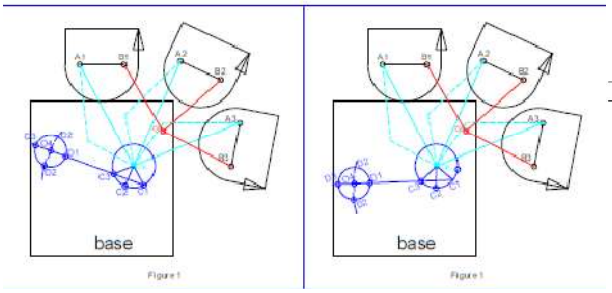


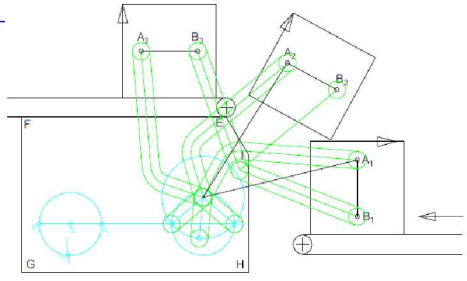


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Adjusting Linkage to Avoid Obstacles

4. Bisect line segment C1 and C2 (or C3 for 3 position synthesis) , and draw a circle of that radius about *02*.
5. Label the two intersections of the circle and *C1C2 or C1C3* extended, *A1* and *A2*.
6. Measure the length of the coupler as *A* to *B1* or *A2* to *B2*.
7. Measure ground length 1, crank length 2, and rocker length 4.
8. Check the Grashof condition. If non-Grashof, redo steps 3 to 8 with O2 further from *04*.





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